



DSA End Module Test

Ques 1 : You are given the starting nodes (start) of two linked lists.

Your task is to check whether both linked lists are exactly the same.

Two linked lists are considered equal if:

- 1.They contain the same number of nodes, and
- 2.Each corresponding node in both lists has the same data value.

If all node values match **and** the lengths are the same, return true (or 1).

If any value is different **or** the lengths are different, return false (or 0).

Example

List 1:

10 → 20 → 30

List 2:

10 → 20 → 30

Output: true (Both lists are identical)

List 1:

10 → 20 → 30

List 2:

10 → 25 → 30

Output: false (Data mismatch)

List 1

10->20->30->null

List 2

10->20->null

Output: False (Count of nodes mismatch)

Que 2 : Doubly Linked List

Write a Java program/function to implement:

insertAtBeginning()

insertAtEnd()

deleteFromBeginning()

display()



Que 3 : Stack

Implement a Stack using an array with the following operations:

push()
pop()
peek()
display()

Also handle:

- Stack Overflow
- Stack Underflow

Que 4: Searching / Sorting

Write a Java program to:

1. Accept an integer array.
2. Sort the array using **Insertion Sort**.
3. Search for a given element using **Binary Search**.
4. Display whether the element is found or not.

Que 5: Graph

Consider the graph:

```
  1
 / \
2   3
/\  \
4 5 6
```

Starting from vertex 1, write:

1. DFS traversal



FirstBit Solutions